

Computer Vision Techniques in Autonomous Vehicles: A Survey

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Abstract:

In today's world where all tasks can be done with the speed of a single click, autonomous vehicles have come into picture which not only ensures safe self-driving but also an effortless driving experience. The demand of such vehicle tends to grow much more in the near future due to which technologies supporting these vehicles are being improvised and researched. Computer Vision enables the computer to see, understand and make observations just as Artificial Intelligence helps the computer to think. Computer Vision is a branch of AI which helps the computer systems to visualize and obtain information through various inputs just as the human eye works in the human body. Computers can gain immense and accurate information through digital images, visuals as well as videos when trained. Computer vision trains algorithms and machines so that they can make self-driving easier. It uses sensor technology to recognize objects and people on the lane. Computer vision algorithms when trained can target the safety of the passengers as well as hassle free driving. The motive of this paper is to introduce the research trends as well as technologies being used for autonomous vehicles using computer vision; we also aim to discuss challenges, advantages as well as future scope of these vehicles.

Keywords- Artificial intelligence, autonomous, computer vision, algorithms

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1. Introduction

The earliest cars were introduced in 1700 which used electricity and steam; later in the next century gas powered cars were first introduced. The evolution of vehicles has been since ancient times introducing new and interesting features and now we have driver less cars which are a boon to humanity as it takes away all the effort by the driver to reach his destinations without having to sit at one place and concentrate. The first idea of driver less vehicles was introduced back in 1939. This concept has reached the next level where they are now known as autonomous vehicles. An autonomous vehicle uses a combination of sensors, Li-DAR, Radar artificial intelligence, radars, and cameras to operate, without any human interference. These sensors observe and inspect the area around the vehicle before it takes any critical decision related to the action to be taken place. After that a localization map is created which can be 2D or 3D. The objective of the map is to recognize the fixed objects around the vehicle. The dynamic objects are removed by removing all Li-DAR points that are found in the area encircling detected dynamic objects. Currently computer vision uses expensive cameras to avoid inconsistency and produce more precise results. Autonomous vehicles are still in an early stage where along with benefits there are certain drawbacks as well; hence these vehicles are sometimes incapable of

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driving through city traffic and bumpy roads. Research is still being made to improve the expectancy of these vehicles and computer vision technology is highly used in these areas.

There are a number of ways in which computer vision can be used in an autonomous vehicle, some of them can be:

- Detecting and classifying objects using Li-DAR sensors and cameras. By classifying static and dynamic objects the vehicle can decide to move accurately and quickly. It will help avoid collisions and mishaps.
- Creation of 3D maps can help the vehicles to know their surroundings and to recognize obstacles in the lane and to avoid them. This can also help in prediction of accidents and hence ensure the safety of the passengers.
- Collecting large data for training algorithms using cameras and sensors can help the vehicle to predict more accurately and easily the types of actions that are needed to be taken. This data can include anything like traffic signals, potholes, maintenance of roads, etc.

Although the advancement in this field there is a lot to be discovered and improvised hence there are challenges that these vehicles face. Collecting the data for training is one of them. Vehicles without human interference need a lot of training and a variety of deep testing. Instead of using a data set it can be better if real time images are captured while driving and are later used for training purposes. After collecting, labeling of data requires a huge amount of manual work. The labeled data must be precise and accurate to help the algorithm train and produce efficient results. Object detection is yet another challenge as the images need classification and localization. AI in self-driving cars is an unending ocean full of discoveries and rapidly advancing technological turns. Hence, we aim to dive more into these advantages and drawbacks in the paper mainly related to the use of computer vision.

The remaining paper is set as follows, in Section II the working of Autonomous vehicles is elaborated in brief, Section III presents the literature survey Next, in Section IV the companies which use Computer vision in their autonomous vehicles are listed. The areas of application of computer vision are discussed in section V, and then in section VI discusses the analysis of literature survey, section VII discusses the challenges faced and possible solutions are presented by us. Then the final Section VII discusses the future scope and conclusion.

2. Working of Autonomous Vehicles

Self-driving also called automated vehicles say that the vehicle is capable of making its own choices and decisions. Autonomous vehicles depend on a lot of complex algorithms, sensors, maps, AI and machine learning to derive its desired objective. The main motive of combining Computer Vision with Artificial intelligence is to form a visual representation that can perform the job that humans do without humans while making the right decisions. Autonomous vehicles use technology to replace drivers with automatic safety features to traverse roads. Even though the idea of an effortless driving seems comforting to the audience there are a lot of challenges that these vehicles are yet to overcome and hence a completely autonomous vehicle is yet to run on the roads. Tesla being a famous example of a self-driving car, still needs active driver supervision and hence does not make the vehicle completely autonomous. Autonomous cars need the driver to give a destination and the car automatically finds the best route. Li-DAR sensors are then used to measure the exact distance of an object and create a 3D map surrounding the car and the car's position is then checked corresponding to the map. It's mounted on the car's top roof in a 360-degree rotating cylindrical compartment. Li-DAR comprises a transmitter, a reflector, and a receiver. To image objects, it uses laser, ultra ultraviolet,

visible light, or infrared light. The emitter shoots a laser beam, which rebounds off a mirror that rotates at 10 revolutions per minute with the cylindrical housing. Cameras are used as sensors to give the best sensor an accurate visual depiction of an autonomous car's nearby area. The cameras are fixed on every of the four sides to give a 360° view. Ultra wide lenses are used in these vehicles to capture a panoramic view that helps the driver with parking. Radars are also used in these vehicles which work on radio waves. They work at different frequencies correlated to short range radars and medium range radars.

Radar is used majorly because it works with millimeter waves and offers precision. It also works better than other sensors even under low range of vision. The Frequency Modulated Continuous Wave radar offers high resolution and fast pulse emission, hence they are used in autonomous vehicles. The next technology that is used when autonomous vehicles are concerned is ultrasonic sensors. These sensors work by measuring the reflected signals. A transducer in ultrasonic sensors produces vibrations that emit ultrasonic pulses. The range is determined by the frequency of the vibration; they are used widely in parking systems to detect the hurdles behind the vehicle to avoid accidents.

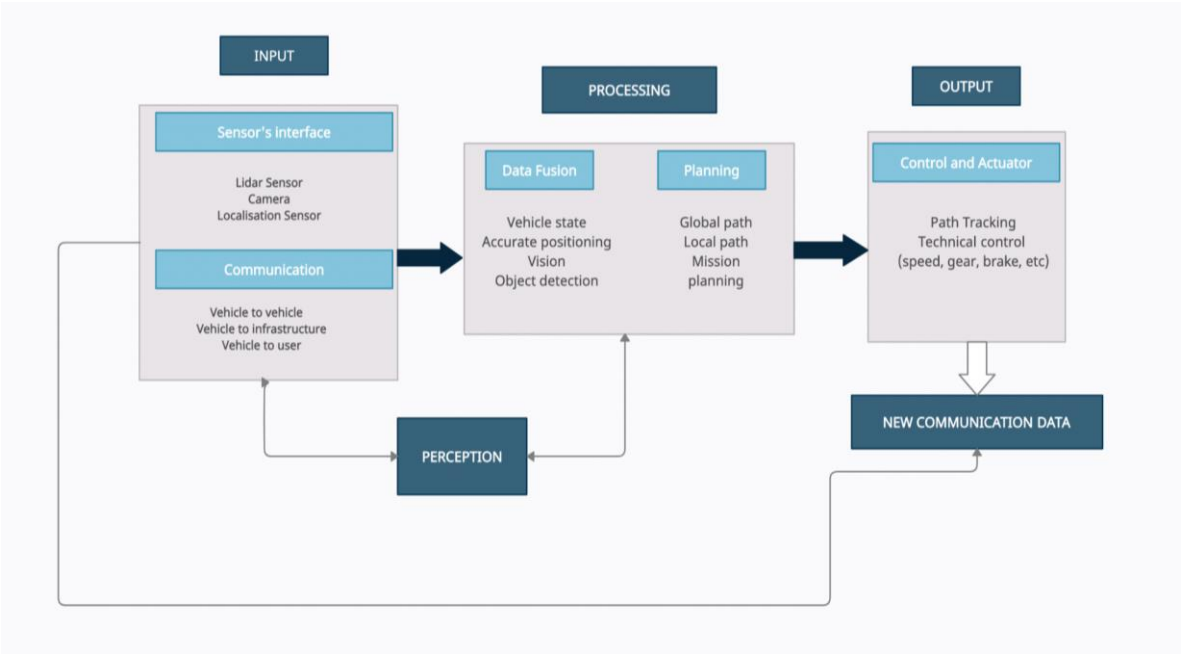


Figure1. System Architecture

Nomenclature	
A	Li-DAR Light Detection and Ranging
B	MPNN Message Passing Neural Network
C	STVV Spatio Temporal Video Volumes
D	COVNET Convolutional Neural Network

2.1 Tables:

Table I. Computer vision in Self- Driving Cars

Sr. no	Year	Methodology	Dataset used	Results	Gaps/ Issues
1.	2021	This paper uses computer vision for extracting features and object detection. It uses CNN using spatial transformer networks and lane detection to improve the accuracy of self- driving vehicles	They have used the German traffic data set containing 50,000 images and divided it into 43 classes. All classes contain traffic signs.	The Le-Net- 5 architecture is compared with feed forward neural network and is about 97% accuracy and the feed forward neural network was 94% accuracy	Le-Net-5 Architecture was unable to perform well on particular traffic signs. The data set did not contain enough examples for all types of signs.
2.	2020	The next paper proposes a support system for autonomous cars for accident detection using dashboard cameras. They have used Mask R-CNN framework	MS COCO dataset is used to evaluate the performance of object detection and image segmentation models, as well as a bespoke data set for lane accidents. The videos were found on the internet. The dash-cam films are 30 frames per second and feature footage from all over the world.	The proposed method has a success rate of 79.05 percent and a false alert rate of 34.44 percent in correctly identifying accidents. The algorithm uses a number of parameters to determine the chance of an accident.	Availability of large data sets for testing and predicting more accurately. The system is also limited by the prediction ability of the Mask R-CNN framework.
3.	2020	This paper proposes a small scale platform which uses computer vision and object detection for autonomous vehicles. They have compared their platform along with the industry level self-driving vehicles	They have used a computer, RGBD camera and an RC car. An open CV software library has been used for path detection, YOLO-v3 for object identification and an Indoor positioning system	The platform shows a comparison between an industry level self-driving vehicle and themselves. It provides access to core technologies and also gives a chance to the researchers to get an idea and improvise autonomous car technologies.	Image based visual servo methods would require more powerful resources for computation. Li-DAR is not used which could provide more data about the environment.
4.	2020	This study employs the canny edge detection approach, with the gradient value of the image used as input calculated using the Sober operator.	There is a lot of noise in the data that is received directly from the camera. To mitigate the effects of these noise concerns, this paper employs a Gaussian filtering approach. Following that, a local contrast enhancement method is employed to improve all or some of the image's attributes.	As a result, the average detection probability is quite close when two choice procedures are applied. It is demonstrated that the method's decision criterion is resilient and capable of achieving high detection accuracy.	Judging whether a driver is in good condition and whether it is drunk driving is also an important factor.

5.	2019	This research employs a semantic scene segmentation method to classify each pixel according to the object class to which it belongs. K-Mean clustering and Expectation Maximization are two algorithms that have been used to deal with picture segmentation issues such as pixel clustering.	They have used an unlabeled data set.	This algorithm showed better results in tasks of bounding box localization as well as semantic segmentation.	On drivers, human positional prediction may be conducted, which now has applications in both semi-autonomous vehicles and ADAS.
6.	2018	This paper uses computer vision and deep learning to train the data. They have also used neural network to classify traffic signs and vehicle detection. Binocular camera and GPU is used to understand the environment	They have used an opacity train data set for lane detection consisting of 39209 images and test data set consisting of 12630 images.	The algorithm of the model was tested on campus as a test using a four wheeled electric car with a binocular camera. The method can be used in assisted driving research and is helpful and reliable for safety of self- driving.	Sensors and high end tools are not used hence this remains as a model for understanding safety in self-driving but cannot be implemented.
7	2018	SSD7 detects autos and walkers with opacity, a data set for self-driving cars, in this paper. As shown with the simulation results, SSD7 was chosen because the network size is small and the throughput is satisfactory.	The data set for this paper was a Unity engine simulation environment. People, vehicles, streets, vegetation, and the air were all present in the virtual environment.	Even though the approach's processing time is longer than segmentation and recognition alone, it still achieves 30 frames a second.	This approach can be used in a more sophisticated environment to train our network from end to end, leading in a more powerful network.

3. Research and Methodology

We will now discuss the findings and research methods applied in various papers and articles which discuss the application of computer vision in autonomous vehicles.

A. *MPNN classifier and HMM Model :*

MPNN classifier is used to categorize monotonous conduct. The MPNN was firstly established by Zaknich et al. The MPNN network is mostly operated on graphs. The main reason behind implementing this model is messaging, updating, reading and reading out functions on different nodes in paragraphs. There are different variations of the MPNN model that share functions but are defined differently.

B. *Object Detection :*

Earlier methods have 2 ways to improvise the task performance: (i) creating a more efficient method for withdrawing regions; (ii) improvising the classifying model. Procedures that utilize one deep learning network have been proposed, involving the YOLO algorithm.

C. *Mean Shift Algorithm :*

Mean shift comes under clustering algorithms along with unsupervised learning. It is also known as the Mode-seeking algorithm. Normal Mean-shift algorithm does not need specifying the number of clusters before only. Clusters are identified by algorithm. Mostly this algorithm is used for doing cluster analysis in computer vision and image processing. This algorithm does not make presumptions therefore it is a non-parametric algorithm.

D. Creating Spatio Temporal Video Volumes (STVVs) :

STVVs are used for detection of computer accidents using deep learning. Detection of anomalies includes 2 steps: First is, deeper functions are trained automatically. Secondly, the outer score for unknown events is calculated. To detect anomalies, STVV around each pixel are represented with the known activity form through clustering.

E. CNN classification :

Convolution Neural Network also said as COVNET. It is used to determine visual imagery. These are proficient in identifying objects from pictures or video. All the different layers of CNN have convolution filters for reducing dimensions. Computer vision works on images and image segmentation is a very important step. There are 2 types of segmentation: Semantic segmentation which categorizes image pixels into n number of classes. Instance segmentation instances of each object are identified.

F. Current Situation :

In autonomous vehicles, computer vision is used to determine various objects. Today most computer vision algorithms get competition from Li-DAR because Li-DAR is able to solve most of the problems which cannot be solved by computer vision algorithms. Computer vision makes autonomous driving safe for people.

G. Data Simulation :

TE Connectivity gives a powerful simulation tool for better speed data connectivity to let a big range of professionals cross the limits of what is possible. Initiate your own creativity in a modern and structured environment. Dive deep right in with our step-by-step method and start creating now. Computer vision utilizes real-time images to develop a 3D map. With the use of 3D maps, autonomous vehicles can decode the driving space for risk-free driving and also opt for alternate routes in criteria of projected collision.

H. Drivers should also be tested :

It is important to do a mouth test to determine the situation of the driver. We can use yawning data sets to train the model and a threshold value can be obtained. The situation of the driver can be determined based on the threshold value. If he is observed yawning multiple times, the state can be stated as sleepy. The methodology proposed uses a framework for accident detection which uses Mask R-CNN for the purpose of object tracking and detection on dash cam videos. The most important criteria for measuring the performance of accident detection framework depends on Accident detection rate and False alarm rate. The dash cam algorithm used in this research uses a small set of variable conditions instead of all the videos captured hence the approach uses a more practical set of data. The system is successful in recognizing 79.05 percent accidents and 34.44 percent false alarm rate under different circumstances. The system depends on the extraction of information from tracked vehicles.

The next research conducted on this topic, focuses on the collaboration with the mono camera and a previously trained model along with multi- regression technique. This previously trained model is re-trained on a local data set with more images taken from a mono camera to detect cars from the road in the working environment. Then the distance of the cars is calculated by depth equation. The research discusses an Anti-

collision algorithm which has various sections such as object detection in which a multi task algorithm is used to obtain information using YOLO which is an object detection tool. The next factor is depth estimation using regression techniques followed by lane and tracking algorithms separately. The lane assignment algorithm uses geometry principles to help the cars stay in their lanes which are determined by each vehicle's boundaries.

4. Computer vision: Areas of Applications in autonomous vehicles

A. Safety in self- driving :

When it comes to safety, recognizing and regulating the environment is widely debated. Some research findings used a binocular image sensor and the NVIDIA Jet son TX2 computing systems with E-net and Open CV for traffic signal recognition, and surrounding vehicles used the MPC technique to make a decision and choose on driving commands in the simulation model and for control in a real-world scenario. Using dual-motor electric vehicles with this method at low speeds aids environmental perception and accordance with the specified. After converting the image into gray scale Gaussian blur is applied to the image followed by canny edge detection and Hough transformation. Road segmentation is done mainly using CNN classification but the drawback that follows is the down sampling which results in loss of edge information. Target object detection and traffic segmentation is done to find out the position and by MPC decision making cars are able to drive slowly through the area.

Approximating the depth is considered a key element in providing the safety of a car and its passengers. Various tools are available which consist of camera radar and Li-DAR which are enabled with multiple camera vision. Estimating the space between camera lenses and the appropriate destination of an object on images taken is necessary for building a stereo vision system. The system uses the concept of triangulation to make space and depth decisions.

B. Detection of Accidents :

Autonomous cars can detect and report accidents in a number of ways, including sending alerts to nearby support centers. Automobile safety is an important aspect in reducing accidents. Computer vision, sensor fusion, localization, optimum path, and control unit are the five critical elements of self-driving cars. A camera is being used to visualize the environment of the vehicle in computer vision. As a consequence, the computer can retrieve specific details and characteristics from these photos. A solution to locate vehicles which uses Gaussian Mixture Model; has been introduced by Hui et al, which tracks vehicles by a mean shift algorithm, but the challenge with this solution is that it fails to perform well under harsh weather and traffic conditions because the parameters are few in number for the solution to perform. Road vehicle accidents have risen in the past few years and hence detecting accidents using computer vision is like a boon, but it also comes with a lot of challenges.

5. Comparative Analysis

The Le-Net -5 architecture outperforms over all other methodologies. The Le-Net -5 model reduces the training time drastically. The Le-Net -5 architecture outperforms over semantic scene segmentation, SSD7 methodology and other techniques. Table I gives the detailed comparison of various state of the arts methodologies.

6. Challenges and Solutions:

Along with all the advantages that autonomous vehicles come with, we cannot overlook the challenges faced by them. Computer vision has made some amazing advancements in the past five years, driven strongly by the increase in performance of parallel computing infrastructure. Especially on roads with Indian traffic it is extremely challenging to let the car take on a complete autonomous mode. The major reasons that these vehicles can face challenge are:

- Lack of proper infrastructure.
- Lack of road network planning.
- Issues with barriers and road blockers.
- Potholes and bad road conditions during rains.
- Allowance of pedestrians, vehicles as well as animals along with non-motorized vehicles altogether on roads.

The vehicles also need extremely intelligent systems to adapt to the quick changing inputs such as the ability to apply brakes, accelerate and steer. A User Interface, Path Optimization Module, Environment Observation and Modeling Module, and Command and Control Module all have a role in the study and implementation of autonomous car movement at a top standard. Forward collision warning, lane departure and lane keeping, Pedestrian and object sensing, Traffic sign recognition, Headlight control, Landscape mapping, Blind Spot Monitoring, and Vehicle to Vehicle Communication are all abilities on another level. One of the biggest concerns is, the sensors ability to navigate in extreme weather, snow blizzards (white-outs), heavy rain, and dust storms and if they can with current sensor technology overcome all these conditions autonomously.

Another fact that has to be accepted is that even though self-driving cars might be better than manual in majority cases but they don't go without accidents. Taking into consideration Tesla's Full Self- Driving, which is the only company that has been close to giving a complete solution. The Tesla comes in two categories - Using the main release software, they are really excellent at self-driving on freeway, but cannot do more than assist the driver in "in-town" areas.

Using the "beta release" software, which is limited to drivers who volunteer for the program and whom Tesla have vetted as having excellent driving skills. The beta software is still making occasional mistakes - around one significant error per hour, hence cannot be completely dependent. The current computer vision technology that is used in self-driving vehicles is endangered to attacks, where hackers exploit the AI's inputs so that it makes errors. For example, researchers have shown they can manipulate a self-driving car from recognizing stop signs by sticking black and white labels on them. Now let's discuss probable solutions:

- Improvement in computer vision technologies used by autonomous vehicles by providing more training data and more testing on real- time complicated environments.
- Providing autonomous vehicles with complementary technologies by equipping computer vision with and creating AI that can understand the environment as precisely as humans can.
- Another solution to solve the issues faced by autonomous vehicles is to designate the roads and surroundings that they run on.

7. Conclusions and Future Scope

Autonomous cars have come a long way since they were first introduced. In this paper we have aimed to discuss how computer vision helps and affects the working of autonomous vehicles as well as the challenges and future scope of this technology. Autonomous vehicles still need to adapt to a lot more situations and the amount of testing and training the data has to be significantly large to avoid errors. Presently, autonomous vehicles require certain human contact and hence are not completely adaptable to changes in the environment, but it is expected that in the coming years, cars will become fully autonomous. Computer vision study has the potential to make meaningful contributions toward the eventual aim of a fully automated vehicle.

The future of autonomous vehicles will be completely connected to humans and objects. The future self-driving cars may also be able to communicate with other similar cars and exchange information. Additionally services such as in-built car Wi-Fi, cloud services, entertainment services can be incorporated to make the passengers feel extreme comfort and luxury. Plenty of technological innovations will take place as the world is introduced to newer technologies. Self-driving cars may also be electric, so they can ensure safety of the environment along with the comfort of autonomy. There are a number of challenges while this comes into actual picture, but the change has already started to take place and we can be sure of looking at such types of fully autonomous cars in the coming years.

The upcoming autonomous cars may also lift people from their stops and transfer them to their desired places on its own. Since there will be no interference of humans the traffic issues can also disperse as the vehicles may have smart road traffic detection and avoidance sensors. Driving licenses also will not be required hence the hassle of driving tests and other documentation will be no longer needed for the drivers. The Internet of Things and communication technology along with Computer Vision techniques will be more advanced in the upcoming years along with new pattern detection, algorithms and sensors as well as efficient collision and object detection algorithms and methods to make the driving experience smoother to its passengers.

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